

A superfactorial obstruction to the proposed lower bound for $B_3(2m + 1)$

Candidate disproof of inequality (27) in arXiv:1210.0835

27 August 2026

Status: candidate counterexample/disproof, likely valid. The lower bound proposed in Question 6 cannot hold: the entire absolute path sum is at most $\binom{m+5}{3}/(4^{m+4}(m!)^2)$, which is superfactorially smaller than $C^m/m!$ for every fixed $C > 0$.

1 Source question

Djakov and Mityagin study weighted walks arising in the spectral analysis of Hill operators [1]. For steps $F = \{-2, +4\}$, $n = 2m + 1$, and exactly three negative steps, their Question 6 on PDF page 5 asks for an asymptotic formula for $B_3(2m + 1)$, “or at least” a positive absolute constant C such that

$$|B_3(2m + 1)| \geq \frac{C^m}{m!}. \quad (1)$$

They note that (1) would imply a non-basis result for an associated antiperiodic Hill operator.

2 Definitions

An admissible walk x starts at $-n$, ends at n , uses steps from $\{-2, +4\}$, and has no internal vertex equal to either endpoint. If its internal vertices are $j(1), \dots, j(\nu)$, its unweighted path factor is

$$h_1(x) = \prod_{t=1}^{\nu} \frac{1}{n^2 - j(t)^2}.$$

Let $X_n^+(3)$ denote the source’s collection of such walks with exactly three -2 steps. Then

$$B_3(2m + 1) = \sum_{x \in X_n^+(3)} h_1(x), \quad n = 2m + 1.$$

The step-balance equation forces every such walk to have $m + 2$ positive steps. Hence it has $m + 5$ total steps and $m + 4$ internal vertices.

3 Proof intuition

No cancellation estimate is needed. Divide positions by two and inspect only the vertices immediately after the first $m + 1$ positive steps. At the p -th selected vertex, at most three negative steps have occurred, so its distance factor from the left endpoint grows at least like p and its distance factor

as we realized in [5, Proposition 25], see formulas (145) - (154). (As a matter of fact, (26) is equivalent to the fundamental identity for Catalan numbers – see in [10] p.117, formulas (14.10) - (14.12).)

But the next level $X_n^+(3)$ gives some hopes.

Question 6. *Find an asymptotic formula for the sequence $B_3(2m+1)$, or at least explain that for some absolute constant*

$$(27) \quad |B_3(2m+1)| \geq C^m/m!.$$

(It is not difficult to see that the upper estimate

$$(28) \quad |B_3(2m+1)| \leq \sum_{x \in X_n^+(3)} |h_1(x)| \leq C_1^m/m!$$

holds.)

The inequality (27) would imply that *the root function system of the Hill operator (1) with potentials of the form*

$$(29) \quad v(x) = ae^{-2ix} + be^{4ix}, \quad a, b \neq 0,$$

subject to antiperiodic boundary conditions does not contain a basis in $L^2([0, \pi])$.

Figure 1: Question 6 and inequalities (27)–(28) on PDF page 5 of arXiv:1210.0835v1.

from the right endpoint grows, in reverse order, at least like $m - p$. Their products contribute two factorials to every path denominator. The number of possible placements of the three negative steps is only cubic in m . Therefore even the absolute sum is of order at most an exponential times $(m!)^{-2}$, ruling out the proposed $(m!)^{-1}$ scale.

4 The obstruction

Theorem 1. *For every integer $m \geq 1$,*

$$|B_3(2m+1)| \leq \sum_{x \in X_n^+(3)} |h_1(x)| \leq \frac{\binom{m+5}{3}}{4^{m+4}(m!)^2}. \quad (2)$$

Proof. Put $n = 2m + 1$. Normalize the positions by writing

$$j(t) = -n + 2r_t.$$

Thus $r_0 = 0$, $r_{m+5} = n$, and the normalized steps are -1 and $+2$. Moreover,

$$n^2 - j(t)^2 = 4r_t(n - r_t). \quad (3)$$

Admissibility says that every internal r_t differs from both 0 and n .

Fix one admissible path. For $p = 1, \dots, m+1$, consider the internal vertex immediately after the p -th $+2$ step. It is internal because the path has $m+2$ positive steps in total. Let $d_p \in \{0, 1, 2, 3\}$ be the number of negative steps that have occurred by then. The selected vertex is

$$r_p = 2p - d_p. \quad (4)$$

For $p = 1$, admissibility gives $|r_1| \geq 1$. For $p \geq 2$,

$$r_p \geq 2p - 3 \geq p - 1.$$

Consequently

$$\prod_{p=1}^{m+1} |r_p| \geq \prod_{p=2}^{m+1} (p-1) = m!. \quad (5)$$

For the other endpoint factor, set $u = m + 1 - p$. From (4),

$$n - r_p = 2u - 1 + d_p.$$

When $u \geq 1$, this is at least $2u - 1 \geq u$. When $u = 0$, it is a nonzero integer by admissibility, hence has absolute value at least one. Therefore

$$\prod_{p=1}^{m+1} |n - r_p| \geq \prod_{u=1}^m u = m!. \quad (6)$$

There are $m + 4$ internal vertices. The three unselected factors $r_t(n - r_t)$ are products of nonzero integers and therefore have absolute value at least one. Combining (3), (5), and (6) yields the uniform single-path estimate

$$|h_1(x)| \leq \frac{1}{4^{m+4}(m!)^2}.$$

Every step sequence is determined by the three positions occupied by its negative steps among $m + 5$ positions. Hence there are at most $\binom{m+5}{3}$ admissible paths. Summing the single-path estimate proves (2). $\square$

Corollary 2. *One has*

$$\lim_{m \rightarrow \infty} (m! |B_3(2m+1)|)^{1/m} = 0.$$

In particular, there is no $C > 0$ for which $|B_3(2m+1)| \geq C^m/m!$ holds for all sufficiently large m .

Proof. By Theorem 1,

$$(m! |B_3(2m+1)|)^{1/m} \leq \frac{\binom{m+5}{3}^{1/m}}{4^{1+4/m}(m!)^{1/m}} \rightarrow 0,$$

using Stirling's formula. The final assertion follows immediately. $\square$

5 Verification and scope

The argument bounds the absolute sum and is therefore independent of any cancellation among paths. The exact rational enumerator in `code/verify_b3.py` checked $m = 1, \dots, 12$, including endpoint admissibility, the signed sum, and the absolute-sum bound. Run it with

```
conda run --no-capture-output -n sandbox
python code/verify_b3.py --max-m 12
```

The finite check is not part of the proof.

This packet disproves only the fallback lower estimate in Question 6. It does not provide an asymptotic formula for $B_3(2m + 1)$, and it does not decide whether the associated antiperiodic root functions form a basis. It shows that inequality (1) cannot be used to obtain the proposed spectral conclusion.

Bounded local-index and arXiv/web searches on 27 August 2026 used the exact $B_3(2m + 1)$ formula, the proposed $C^m/m!$ scale, the paper title and authors, and the associated two-exponential potential. They found the source and the authors' nearby follow-up arXiv:1210.3907, whose stated antiperiodic range does not include the present $r = 1, s = 2$ case, but no later treatment of this exact bound. Novelty confidence is moderate because the search was bounded and the obstruction is elementary.

References

- [1] P. Djakov and B. Mityagin, *Combinatorial identities related to eigenfunction decompositions of Hill operators: Open Questions*, Integral Equations Operator Theory **75** (2013), 7–12; arXiv:1210.0835v1, <https://arxiv.org/abs/1210.0835>.